هاكاثون الصّنَاعة
Industrial Hackathon

SIDF
صندوق التنمية الصناعية السعودي



Coca-Cola®

Supporting Document

راعي بلاتيني

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سبكيم Sipchem EXCELLENCE everywhere

الشركاء

بنك الرياض Riyad Bank

Truck load capacity

- All trucks are side loading only.
- 21% only with tailgate, where can be load/offloaded from back door using forklift or hand jack.
- Each truck can take maximum of 8 Pallets only. 4% can accommodate 12 Pallet.
- Maximum load 9,000 KG













Data on order volumes and the number of trips per day and per week

#	Number of Customers	Total Cases	Loaded Kg.	Truck Capacity	Load %	Truck
1	24	1,180	8,209	1,200	139%	08 Bay
2	23	962	7,745	850	113%	08 Bay
3	23	1,020	7,788	850	120%	08 Bay
4	24	861	7,123	850	101%	08 Bay
5	7	450	4,080	850	53%	04 Bay
6	17	1,457	8,404	1,200	171%	08 Bay
7	15	848	8,181	850	100%	08 Bay
8	18	1,302	8,295	1,200	153%	08 Bay
9	22	972	7,812	850	114%	08 Bay
10	20	1,053	8,273	850	124%	08 Bay
11	19	990	8,563	850	116%	08 Bay
12	23	884	8,215	850	104%	08 Bay
13	11	762	6,744	850	90%	08 Bay
14	12	1,125	8,920	850	132%	08 Bay
15	15	1,026	8,371	850	121%	08 Bay
16	19	968	8,798	850	114%	08 TG
17	19	1,255	9,000	850	148%	08 Bay
18	17	1,271	8,724	850	150%	08 Bay
19	19	1,052	8,812	850	124%	08 Bay
20	22	1,083	8,474	850	127%	08 Bay
21	19	1,018	8,879	850	120%	08 Bay
22	17	1,210	8,066	850	142%	08 Bay
23	16	1,096	8,662	850	129%	08 Bay
24	7	799	7,546	850	94%	08 TG
25	7	1,336	8,086	850	157%	08 TG
26	8	1,255	7,903	850	148%	08 TG
27	9	891	6,209	850	105%	08 TG
28	9	1,237	7,790	850	146%	08 Bay
29	7	1,202	7,378	850	141%	08 Bay
30	7	1,394	7,329	1,200	164%	08 TG
31	7	1,300	6,972	850	153%	08 Bay
32	8	534	3,296	850	63%	08 TG
33	8	1,297	7,480	850	153%	08 Bay
34	8	1,233	7,352	850	145%	08 Bay
35	24	254	4,207	650	39%	06 Bay
36	23	192	2,983	650	30%	06 Bay
	553	36,769		30,200	122%	

- One Day example of the load summery showing each truck load (Number of Customers, Total Cases, Truck Capacity, type of the truck)
- TG = Truck with Tailgate
- Pay = how many pallets can truck accommodate.

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Data on order volumes and the number of trips per day and per week

						DELIVER			PICK UP		
TRIP	Orders	KM	START WORK TIME	END WORK TIME	DURATION	KG	CBS	CASES	KG	CBS	CASES
RY632	12	124.89	07:55	15:07	7:12	7890	832	1083	0	0	0

NR	CUST NO	NAME	ADDRESS	ZIP CODE	TOWN	DELIVERY WINDOWS	KG	CBS	CASES	ARRIVAL
1	30	Riyadh Center Region				00:01-23:59	7890	832	1083	07:55
2	3073420	DOOS MUSEEF	MALIK IBN OMER MUSEEF		RIYADH	09:00-17:00	443	41	75	09:00
3	3074778	AL SAEED REST AND KITCHENS CO	IMAM SAUD EXIT9		RIYADH	09:00-20:00	8	1	1	09:19
4	3062331	ASWAQ BAWASIQ	ABI BAKR SIDDIQUE		RIYADH	06:30-14:00	534	59	69	09:34
5	3065657	NAKHLA SASSCO COMPANY 11301	OTHMAN BN AFFAN RD		RIYADH	07:00-21:00	541	49	77	09:45
6	3074015	SORRAH COMPANY	OTHMAN BIN AFFAN		RIYADH	09:00-20:00	2694	273	244	10:01
7	3073755	LAGATE COMPANY	UTHMAN BIN AFFAN RD		RIYADH	07:00-21:00	206	21	25	11:25
8	3064161	TAM HI AL FALAH	SHARA IBRAHIM AL DAHAN		RIYADH	07:00-21:00	156	14	22	11:37
9	3076229	MATAM ABDELAZIZ AL SEBAEE	AL ONAZY ST		RIYADH	09:00-20:00	337	36	43	11:48
10	3076229	MATAM ABDELAZIZ AL SEBAEE	AL ONAZY ST		RIYADH	09:00-20:00	517	53	62	12:04
11	3071242	ABRAJ AL RIYADH TRADING CO	OTHMAN BIN AFFAN ROAD 12		RIYADH	07:00-21:00	247	27	72	12:27
12	3070963	DARK STORES - NARJIS	ABI BAKR SADDIQ		RIYADH	08:00-14:00	1895	227	362	12:54
13	3074491	FULL STOP NARJIS	ANAS BIN MALIK		RIYADH	09:00-20:00	312	31	31	13:48
14	30	Riyadh Center Region				00:01-23:59				15:07

UNLOADING TIME	DRIVING TIME	DISTANCE	ODT	COMMENT3
00:00	01:05	50.08		
00:16	00:03	0.85	1	https://www.google.com/maps/place/24.754932,046.67642
00:07	00:08	5.13	1	https://www.google.com/maps/place/24.752062,046.67465
00:00	00:11	3.86	1	https://www.google.com/maps/place/24.782168,046.68341
00:14	00:02	0.33	1	https://www.google.com/maps/place/24.798126,046.69732
01:22	00:02	0.68	1	https://www.google.com/maps/place/24.795443,046.69854
00:08	00:04	1.64	1	https://www.google.com/maps/place/24.790361,046.70091
00:07	00:04	1.15	1	https://www.google.com/maps/place/24.801430,046.69948
00:16	00:00	0.00	1	https://www.google.com/maps/place/24.795738,046.70412
00:14	00:09	3.21	1	https://www.google.com/maps/place/24.795738,046.70412
00:08	00:19	6.35	1	https://www.google.com/maps/place/24.800488,046.69241
00:51	00:03	0.63	1	https://www.google.com/maps/place/24.837889,046.68775
00:16	01:03	50.98	1	https://www.google.com/maps/place/24.835790,046.68200
00:00	00:00	0.00		

One Driver/ Truck load details

- Outlet name
- Receiving window time
- Outlet order in cases, KG and Cubes.
- Location

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Current average loading and unloading times

Key Account Delivery window Time

- Between 6:00 AM up to 2:00 PM
- Priority for diary products, in receiving with the customer.
- Average waiting/ delivery time with one customer is 2 Hours.
- Average load for 1 customer is 100 Cases

Restaurants

- Between 10:00 AM up to 05:00 PM
- Delivery time Avg. for one customer is 40 min
- Average load for 1 customer is 20 Cases

Dark stores

- Between 07:00 AM up to 05:00 PM
- Delivery time Avg. for one customer is 2 Hours
- Average load for 1 customer is 200 Cases

Other Presells

- Between 07:00 AM up to 10:00 PM
- Delivery time Avg. for one customer is 40 min
- Average load for 1 customer is 70 Cases

Current estimated logistics cost rates

Delivery Cost

Type	Cost/ day
Truck+Fuel	250 SAR
Driver	100 SAR
Helper x2	150 SAR
Total	442 SAR

Storage Cost

Type	Cost/ day/ Case
Storage	0.50 SAR
Handling	0.25 SAR
Returns	0.50 SAR
Total	1.25 SAR

Some of the products







Packaging

Different sizes of product required different type of packaging.
Where all product must be in good sealed to avoid leakage.

Bottle x Sizes/ml	Cases in Pallets
6x150	448
24x250	392
24x320	130
24x330	104
12x350	132
12x500	132
6x1.0 L	150
6x1.5 L	100
6x2.2 L	72
6x2.25 L	72

SKU	Bottles Types
Coca-cola	Can - PET - NRB
Sprite	Can - PET - NRB
Fanta	Can - PET - NRB
Monester	Can
Arwa	PET

Packaging Types	Comment
Can	Aluminum
PET	Plastic bottle, usually for +500 ml SKU
NRB	Non returnable broducts - glass
Arwa	all plastic bottles.

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Solutions that have been tested previously











